# IoT LAB (EC2601-1)

## A Project Report on

# Intruder Detection System Using Face Recognition

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# CERTIFICATE

This is to certify that ***B Pranav Bhaskar Nayak (NNM23EC036), Bhargavi Shetty (NNM23EC037), Bhavish U R (NNM23EC038), Bhoomika Paradkar (NNM23EC039)***, are bonafide students of N.M.A.M. Institute of Technology, Nitte, who have submitted a project report entitled "**Intruder Detection System Using Face Recognition**" as part of the $6^{th}$ semester IoT Lab, in partial fulfillment of the requirements for the award of Bachelor of Engineering Degree in Electronics and Communication Engineering during the year 2025-2026.

| **Name of the Examiners** | **Signature with date** |
|---|---|
| **1.**____________________ | ____________________ |
| **2.**____________________ | ____________________ |

# ABSTRACT

The Intruder Detection System Using Face Recognition is an intelligent security solution designed to enhance safety using real-time monitoring and face recognition technology. The system utilizes a webcam to capture live video, which is processed using OpenCV for face detection and DeepFace for face recognition. A dataset of authorized users is maintained, and the system compares detected faces with stored images to determine whether the person is authorized or an intruder.

When an authorized individual is detected, the system communicates with an ESP32 microcontroller to activate a servo motor for door access and turn on a green LED as an indication. In contrast, if an unknown person is detected, the system triggers multiple alert mechanisms including buzzer activation, red LED indication, image capture, and email notification with the captured image attached.

The system demonstrates the effective integration of computer vision and IoT technologies to provide a reliable, automated, and real-time security solution. It is cost-effective and suitable for applications such as home security, office monitoring, and restricted access areas.

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# Chapter 1

# INTRODUCTION

## 1.1 Background

Security systems are essential in modern society to protect homes and organizations. Traditional systems rely on passwords or keys, which can be lost or compromised. With advancements in computer vision and IoT, intelligent systems using facial recognition provide a more secure solution.

## 1.2 Problem Statement

Conventional security systems lack real-time monitoring and intelligent decision-making. There is a need for an automated system that can detect intruders, verify identities, and trigger appropriate actions.

## 1.3 Objectives

1. To develop a face recognition-based security system.
2. To detect and differentiate between authorized and unauthorized users.
3. To send alerts and control hardware using ESP32.

## 1.4 Scope

The Intruder Detection System Using Face Recognition is designed to provide a smart and automated security solution for small-scale applications such as homes, laboratories, and offices. The system focuses on real-time face detection and recognition to identify authorized users and detect intruders efficiently.

The scope of the project includes integration of computer vision techniques using OpenCV and DeepFace with IoT-based hardware components like ESP32, servo motor, LEDs, and buzzer. It enables real-time monitoring through a webcam and provides immediate alerts through email notifications and audio feedback.

The system is limited to basic face recognition functionality and depends on the quality of the dataset and lighting conditions for accurate performance. It does not include advanced features such as multi-user authentication, cloud storage, mobile application interface, or large-scale surveillance.

Future scope includes enhancing accuracy using advanced deep learning models, integrating mobile applications for remote access, implementing cloud-based data storage, and expanding the system for large-scale security applications such as smart cities and industrial monitoring.

# Chapter 2

# LITERATURE REVIEW

Existing intruder detection systems have evolved from basic alarm-based systems to advanced intelligent systems using computer vision and IoT. Traditional systems relied on motion sensors and CCTV monitoring, which often resulted in false alarms and required human supervision. Modern systems use cameras and image processing techniques to automatically detect and identify intruders.

Many systems are developed using software tools like OpenCV, which provides efficient algorithms for face detection, image processing, and video analysis. These systems capture real-time video, detect human faces, and compare them with stored datasets to identify authorized and unauthorized users. When an intruder is detected, alerts such as email notifications, buzzer activation, or mobile messages are triggered .

Python has been widely used in developing intruder detection systems due to its simplicity and powerful libraries such as OpenCV, NumPy, and DeepFace. OpenCV alone provides thousands of optimized algorithms for image processing and object detection, making it suitable for real-time applications. Python also allows easy integration with hardware devices like ESP32 through serial communication, enabling automation.

In educational programming environments, Python is highly preferred because it is easy to learn, readable, and supports rapid development. Students can quickly build practical applications like face recognition-based security systems without requiring deep knowledge of complex programming languages. Thus, Python plays a significant role in developing efficient, low-cost, and real-time intruder detection systems.

## 2.1 Survey

1. **"Survey of Best Home Intruder Detection by Face Recognition and Touch Sensor"**
   **Authors:** Abinaya Shri and S.P. Chokkalingam
   In this paper,a dual-trigger intruder detection system using motion and touch sensors along with a camera module. System integrates Firebase and an Android application for real-time monitoring. Alerts are sent via both internet notifications and GSM-based SMS.

2. **"Facial-based Intrusion Detection System with Deep Learning in Embedded Devices"**
   **Authors:** Giuseppe Amato, Fabio Carrara, Fabrizio Falchi, Claudio Gennaro
   This work focuses on implementing deep learning-based face recognition on embedded devices like Raspberry Pi. It uses the ResNet-50 model for feature extraction and

k-Nearest Neighbor (kNN) for classification. The system achieves efficient real-time recognition without requiring GPU support.

3. **"Face Recognition using Deep Learning for Real-time Intruder Detection"**
   **Authors:** Presana Periasamy Pillai and Kalaivani Chellappan
   This paper presents a real-time intrusion detection system using ResNet-34 and Histogram of Oriented Gradients (HOG). Alerts are sent via WhatsApp API and a buzzer is activated for immediate response.

4. **"IoT Based Home Intruder Alerting System"**
   **Authors:** J. Jebin Gerald, Immanuel B., and S. Poornapushpakala
   This system uses Arduino UNO and machine learning algorithms such as XGBoost for intrusion detection. It provides SMS alerts via GSM module and supports two-way voice communication, allowing users to interact remotely with visitors or intruder.

5. **"Implementation of Low Cost IoT Based Intruder Detection System by Face Recognition using Machine Learning"**
   **Authors:** G. Mallikharjuna Rao, Haseena Palle, Pragna Dasari, and Shivani Jannaikode
   This paper presents a low-cost system using Haar Cascade and Local Binary Pattern (LBP) algorithms for face detection. The system is optimized for speed and sends email alerts using SMTP when an intruder is detected. It also activates a buzzer for immediate warning.

# Chapter 3

# METHODOLOGY

## 3.1 Block Diagram

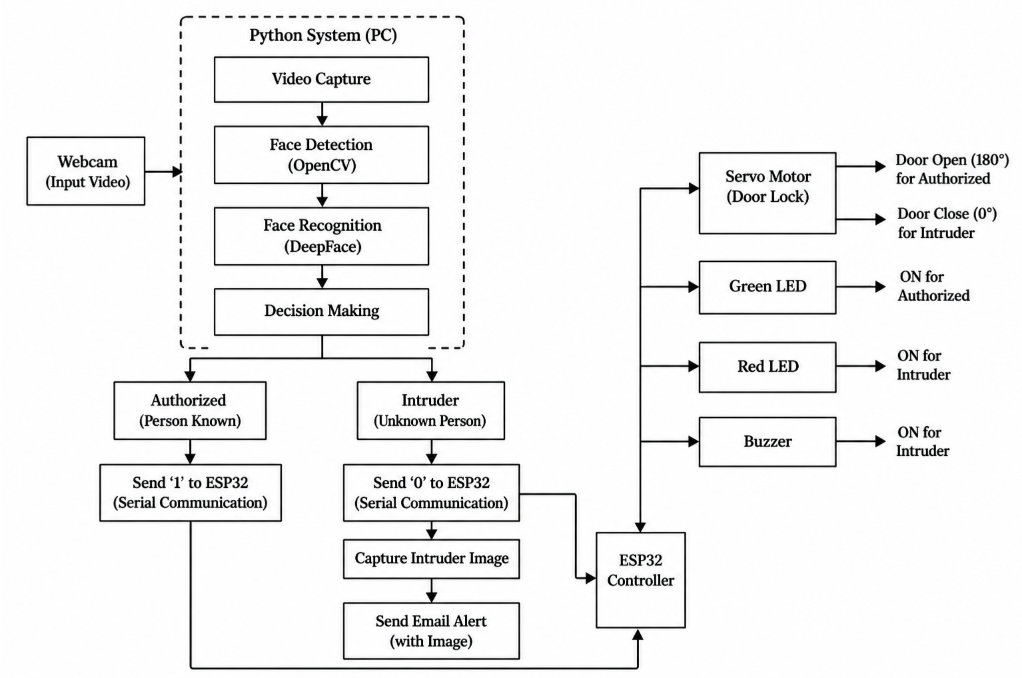


Figure 3.1: System Block Diagram

## 3.2 flow chart

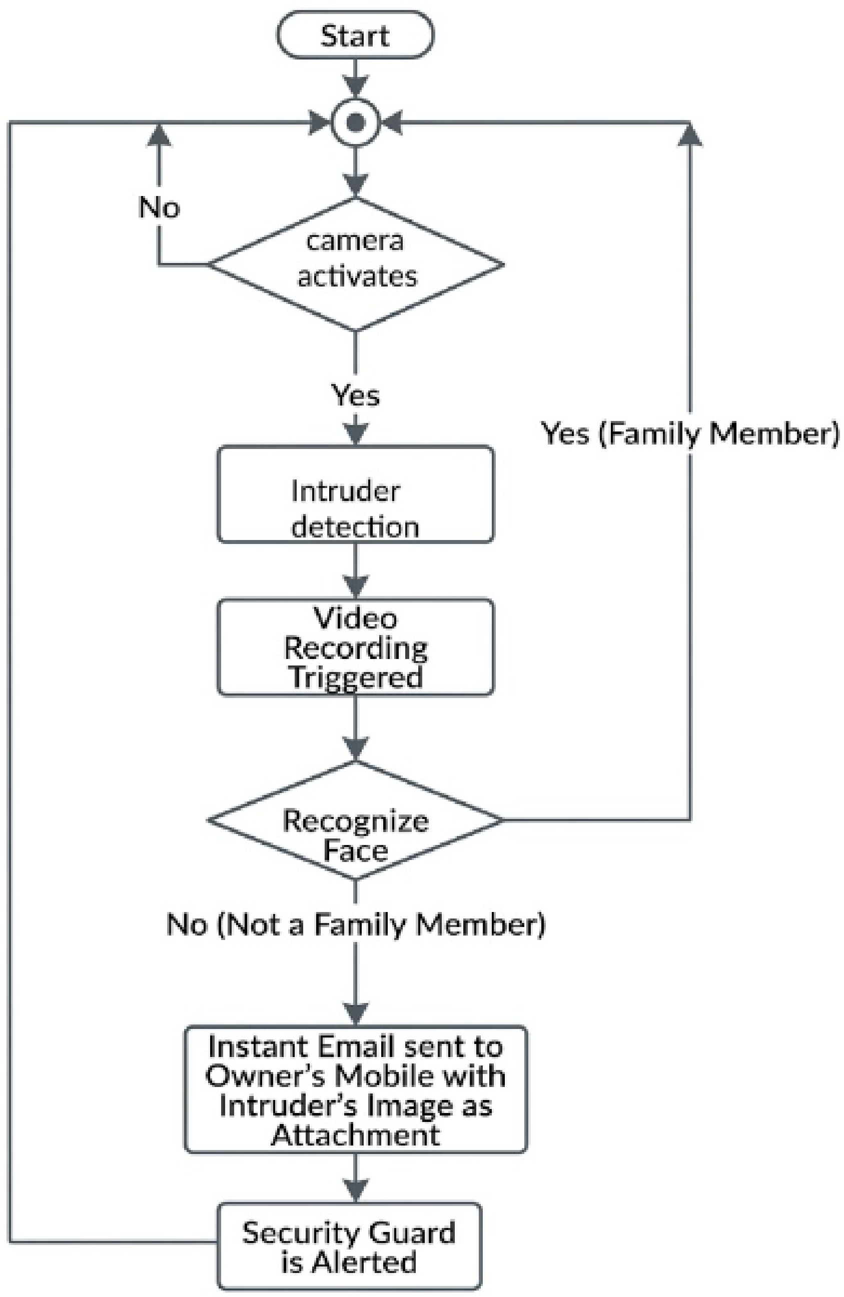


Figure 3.2: Flow Chart for Face detection

## 3.3 Working Principle

- Webcam captures real-time video
- OpenCV detects faces
- DeepFace verifies identity
- If authorized → Door opens using ESP32 + Servo + Green led
- If intruder → Buzzer ON + Email alert sent + Red led

# Chapter 4

# HARDWARE AND SOFTWARE REQUIREMENTS

## 4.1 Hardware Requirements

The following hardware components are used to implement the Intruder Detection System Using Face Recognition:

- **ESP32 Microcontroller:** Used to control hardware components and communicate with the computer via serial communication.
- **Webcam:** Captures real-time video input for face detection and recognition.
- **Servo Motor:** Used to simulate door locking and unlocking mechanism.
- **Buzzer:** Generates an alert sound when an intruder is detected.
- **LEDs (Red and Green):** Indicate system status (authorized or intruder).
- **Connecting Wires and Breadboard:** Used for circuit connections.

### 4.1.1 ESP32

ESP32 is a low-cost, low-power microcontroller with built-in Wi-Fi and Bluetooth capabilities. It is used in this project to control the hardware components such as LEDs, buzzer, and servo motor. The ESP32 receives commands from the Python program through serial communication and performs actions accordingly. It plays a key role in enabling IoT-based control and automation.

## 4.2 Software Requirements

The following software tools and libraries are used in the development of the system:

- **Python:** Used for implementing face detection, recognition, and system logic.
- **OpenCV:** Used for image processing and face detection.
- **DeepFace:** Used for face recognition using deep learning models.

- **PySerial:** Used for serial communication between Python and ESP32.
- **SMTP (smtplib):** Used to send email alerts.

### 4.2.1 Arduino IDE

Arduino IDE is used to program the ESP32 microcontroller. It provides a simple interface for writing, compiling, and uploading code to the ESP32. The IDE supports serial communication, which is used in this project to receive commands from the Python program and control hardware components such as LEDs, buzzer, and servo motor.

# Chapter 5

# RESULTS AND DISCUSSION

## 5.1 Presentation of Results

The system successfully detects faces and identifies whether the person is authorized or an intruder. When an authorized person is detected, the servo motor opens the door and green LED turns ON.

When an intruder is detected:

- Red LED turns ON
- Buzzer is activated
- Image is captured
- Email alert is sent with image attachment

Figure 5.1 shows the detection of an authorized person by the system. The face recognition module correctly identified the individual from the dataset, and the system granted access. As a result, the green LED was turned ON, indicating successful authentication. This demonstrates the proper integration of software-based face recognition with hardware control using ESP32.

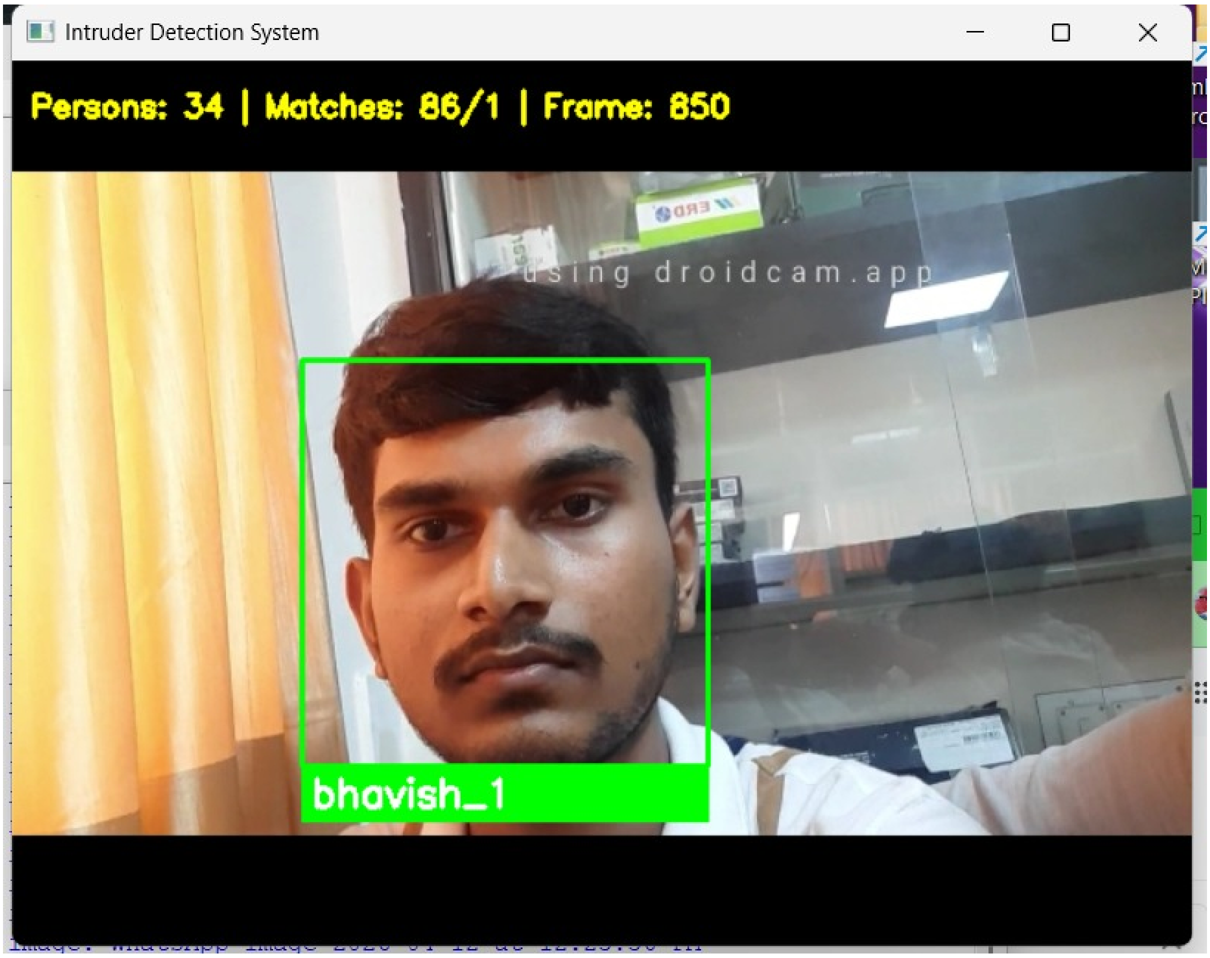


Figure 5.1: System Output

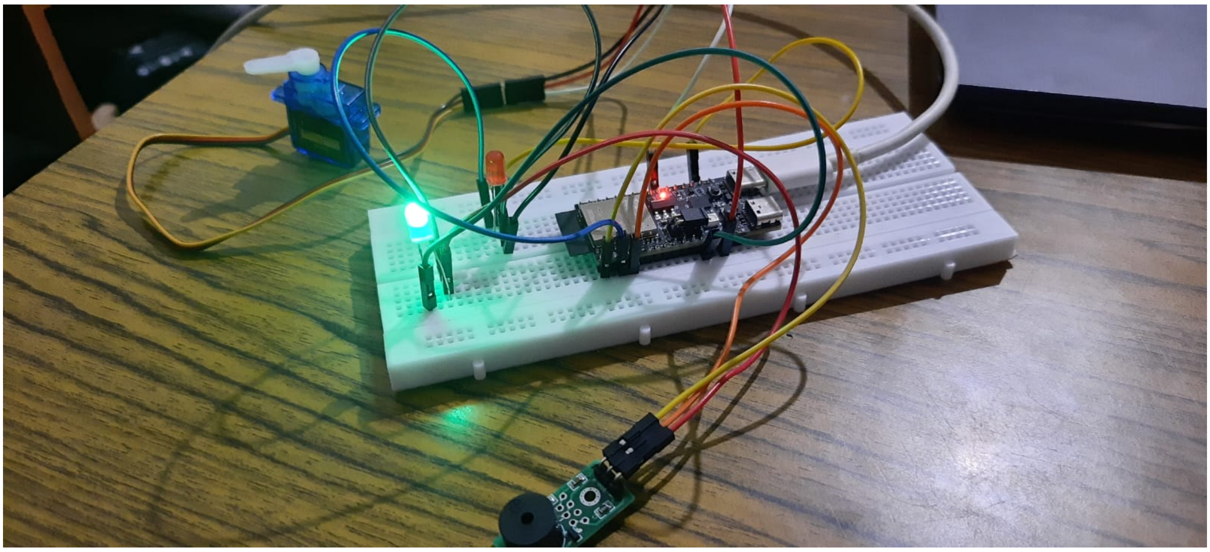

Figure 5.2: Led Output

## 5.2 Analysis of Results

The proposed Intruder Detection System Using Face Recognition demonstrates efficient performance under real-time operating conditions. The system is capable of detecting and recognizing faces accurately using computer vision techniques. The face recognition accuracy primarily depends on factors such as the quality of the dataset, variations in lighting conditions, and camera resolution. Under well-lit conditions and with a properly trained dataset, the system achieves reliable and consistent results in distinguishing between authorized users and intruders.

The integration of IoT components, particularly the ESP32 microcontroller, significantly enhances the system's automation and response time. The communication between the Python-based software and the ESP32 hardware is seamless, enabling immediate actions such as activating LEDs, buzzer alerts, and controlling the servo motor for door access. The green LED indication for authorized users and red LED with buzzer for intruders clearly demonstrate the correct functioning of the hardware interface.

The email alert feature plays a crucial role in enabling remote monitoring. Upon detecting an intruder, the system captures an image and sends it via email, ensuring that the user is instantly notified even when not physically present. This enhances the overall reliability and effectiveness of the system as a security solution.

However, certain limitations were observed during testing. The performance of the system may degrade in low-light environments or when faces are partially occluded. Additionally, recognition accuracy depends on the diversity and quality of images in the dataset. Despite these limitations, the system performs satisfactorily for small-scale applications and demonstrates the practical implementation of IoT and computer vision technologies in real-time security systems.

# CONCLUSION

The Intruder Detection System Using Face Recognition successfully achieves its objective of providing a smart and automated security solution using face recognition and IoT technologies. The system is capable of accurately detecting and distinguishing between authorized users and intruders in real time, ensuring enhanced security and reduced human intervention.

The integration of OpenCV and DeepFace enables efficient face detection and recognition, while the ESP32 microcontroller ensures seamless control of hardware components such as LEDs, buzzer, and servo motor. The implementation of email alerts and image capture further strengthens the system by enabling remote monitoring and quick response to security breaches.

Although the system performs effectively under normal conditions, factors such as lighting variations and dataset quality can influence recognition accuracy. Future enhancements may include the use of advanced deep learning models for improved accuracy, mobile application integration for better user interaction, and cloud-based storage for data management.

Overall, the project demonstrates the practical application of computer vision and IoT in developing intelligent security systems and provides a strong foundation for further research and development in this domain.

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